



DATABASE-SYSTEMS

Lecture - 4

By

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Relational Database Models

Once Again Back to Relational Database Models

Revision (Core Concepts)

As we know it was combination of following:

Tables

(Relations)
(Entities)

Rows

(Records)
(Tuples)
(Entity Instance)

Columns

(Characteristics)
(Attributes)

Degree

(Entity to Entity)

Cardinality

(Record to Record)

Domain

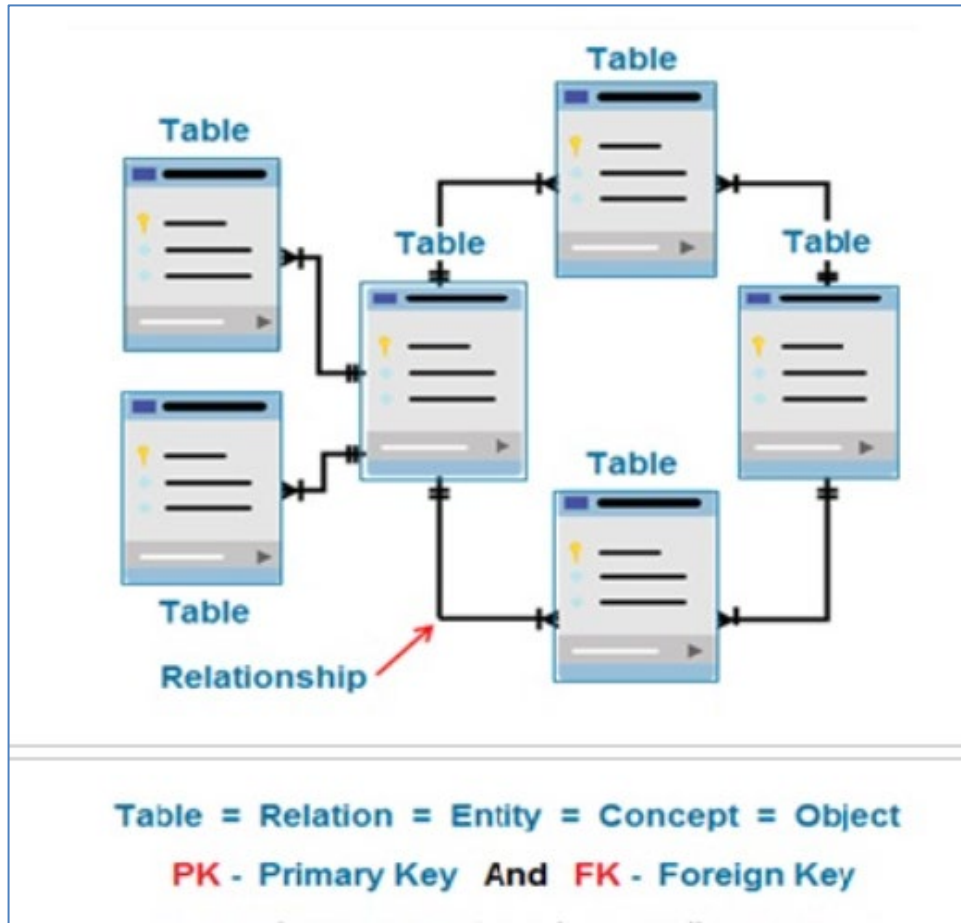
(Possible values of attribute)



Relational Database Models

Revision (Core Concepts)

The structure was some thing like this:



Column Attribute

StuID	StuName	StuAge	StuClass	StuSection
1001	Alex	15	10	B
1002	Maria	14	11	A
1003	Maya	14	9	A
1004	Bob	16	11	C
1005	Newton	14	10	D
1006	Sanjay	15	10	B

Record Tuple →

← Row

Table / Entity / Relation

A **Table** Represents a Database **Entity**

Table **Row** is referred as **Records** Or **Tuple**

A Table **Column** Represents an **Attribute**

Relational Database Models

THIS TIME GO MORE DEEP INTO IT

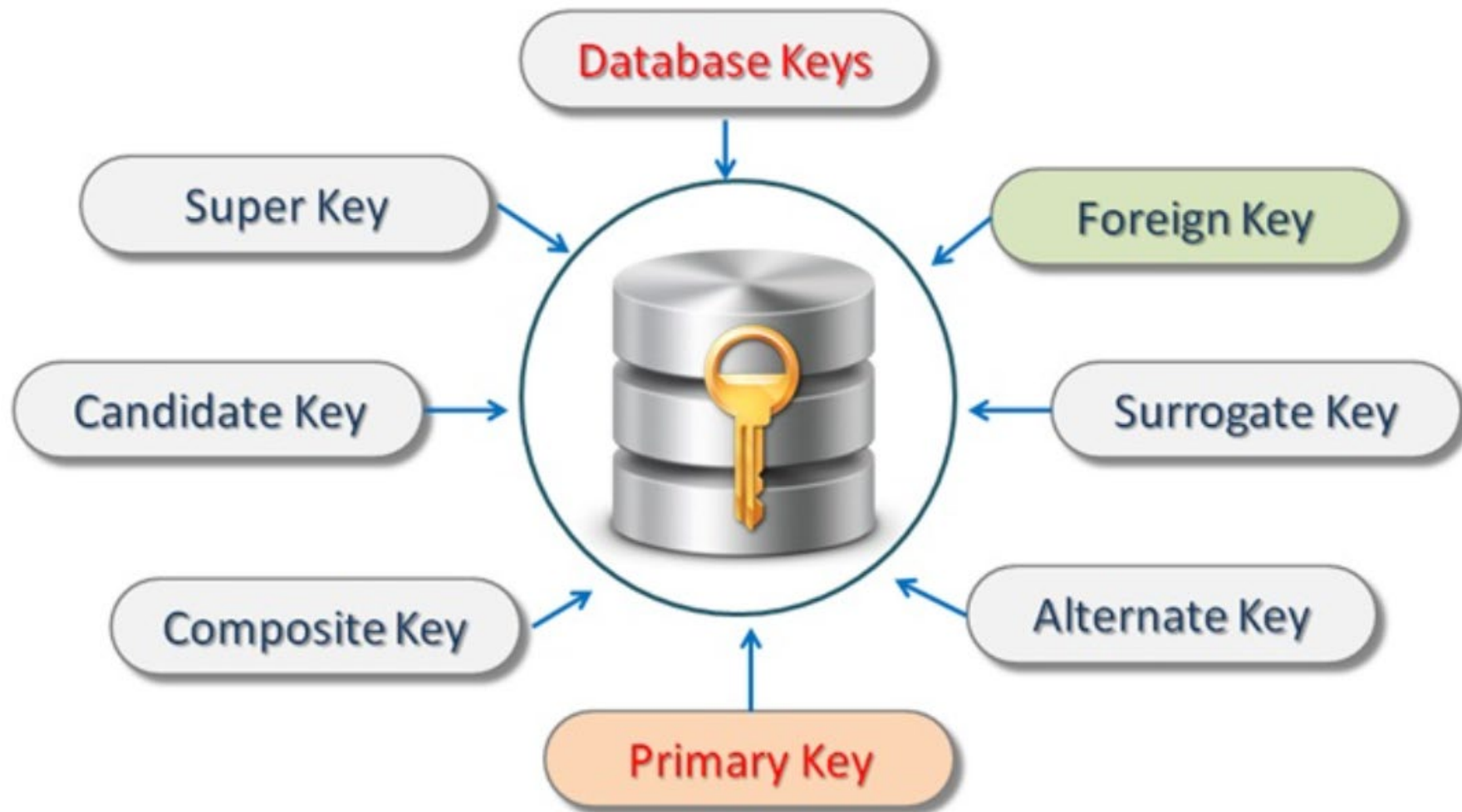
Concept of KEYS in Relational Databases

- 1 A key in a table/relation might be an attribute or set of attributes.
- 2 It helps in uniquely identifying a tuple/row/record in a table/relation.
- 3 These are defined to sequence and access/store the data quickly.
- 4 Keys also help in establishing relationship between different tables/columns.



Relational Database Models

Types of Keys in Relational Database Models



Relational Database Models

Why Do we need keys in Database

- 1 To reduce duplication/redundancy (*Remember: Redundancy is Enemy*).
- 2 To enhance consistency.
- 3 To give/assign a reference value to each record in database w.r.t to columns.

		Database Table						
Row 1	Record 1							
Row 2	Record 2							
Row 3	Record 3							
Row 4	Record 4			→	Data Field			
Row 5	Record 5				↑			
Row 6	Record 6							
Row 7	Record 7							
		Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7
		Attribute 1	Attribute 2	Attribute 3	Attribute 4	Attribute 5	Attribute 6	Attribute 7

Relational Database Models

Why Do we need keys in Database

“In the table each **column** have **unique name** but the table **rows** does not have **unique name**. Therefore we **need database key** so that **each row** in table can be **uniquely identified**”.

Student Table

Student ID	First Name	SSN	Course ID
1001	S1	CA245	CS 1
1002	S2	WA465	CS 2
1003	S3	AR365	CS 2
1004	S4	WA764	CS 1
1005	S5	CA453	CS 3
1006	S6	CA924	CS 1

Course Table

Course ID	Course	Duration	Instructor ID
CS 1	Java	60 Days	PRG 1
CS 2	Python	45 Days	PRG 2
CS 3	C++	55 Days	PRG 3

Instructor Table

Instructor ID	First Name	Contact
PRG 1	Peter	617145
PRG 2	Smith	100023
PRG 3	Rick	454510

Student ID Is **Primary Key** For Student Table

Course ID Is **Primary Key** For Course Table And **Foreign Key** In Student Table

Instructor ID Is **Primary Key** For Instructor Table And **Foreign Key** In Course Table



Relational Database Models

Why Do we need keys in Database (Example)

A **table key** can be a **unique single attribute** such as student ID", employee ID, SSN (Social Security Number) that has an unique value for each row in a table.

What if a table don't have any unique attribute in Database

If the table doesn't contains a unique attribute that can be used as database key then the **combination of two or more keys can be used** to define the database key for that table.

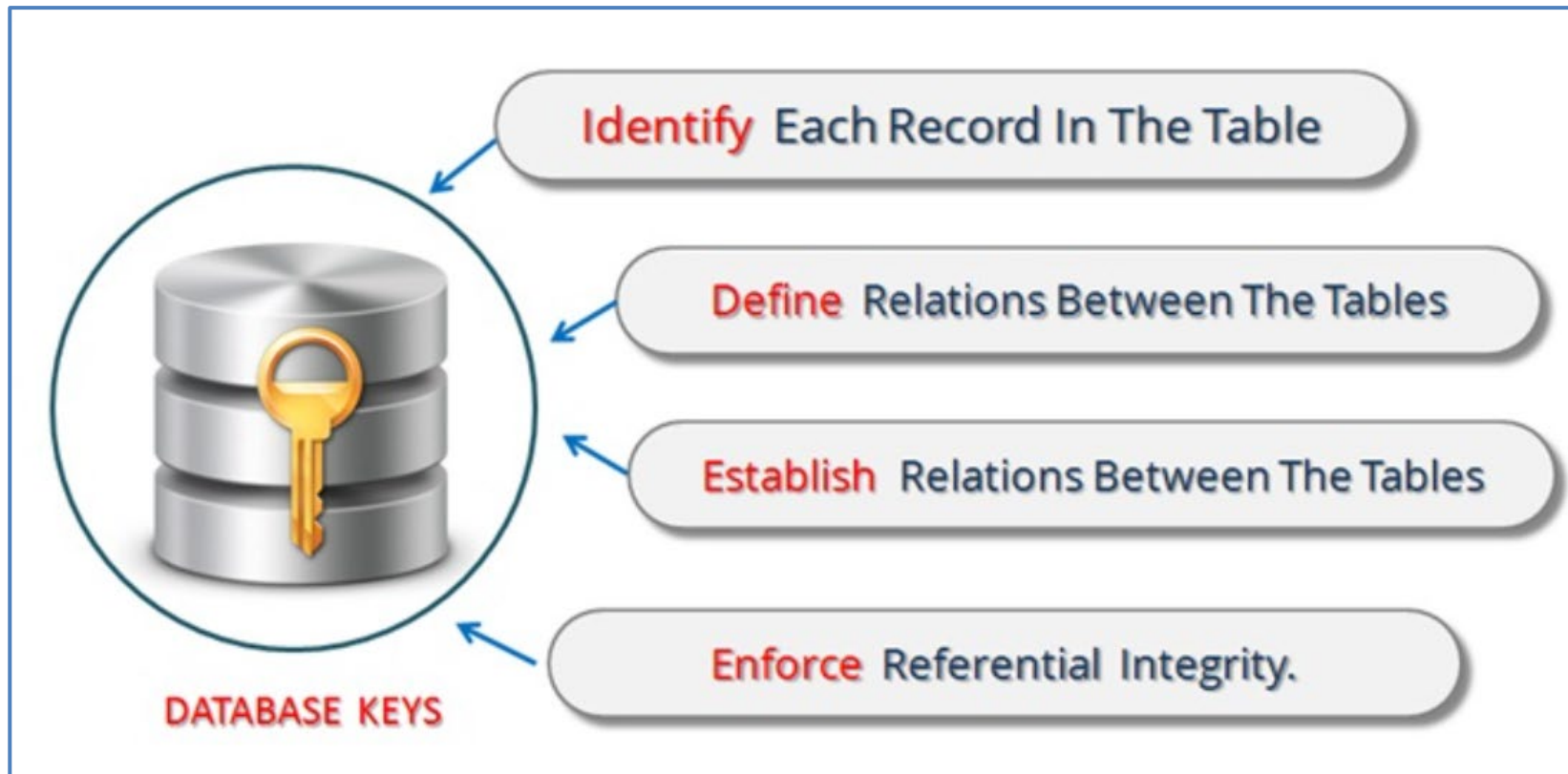
How do we apply a key in Database

In database, system designers/ database designers **first check all the entities**, then they **analyze all attributes** and **find the one with unique values** to use it as a key attribute.

Relational Database Models

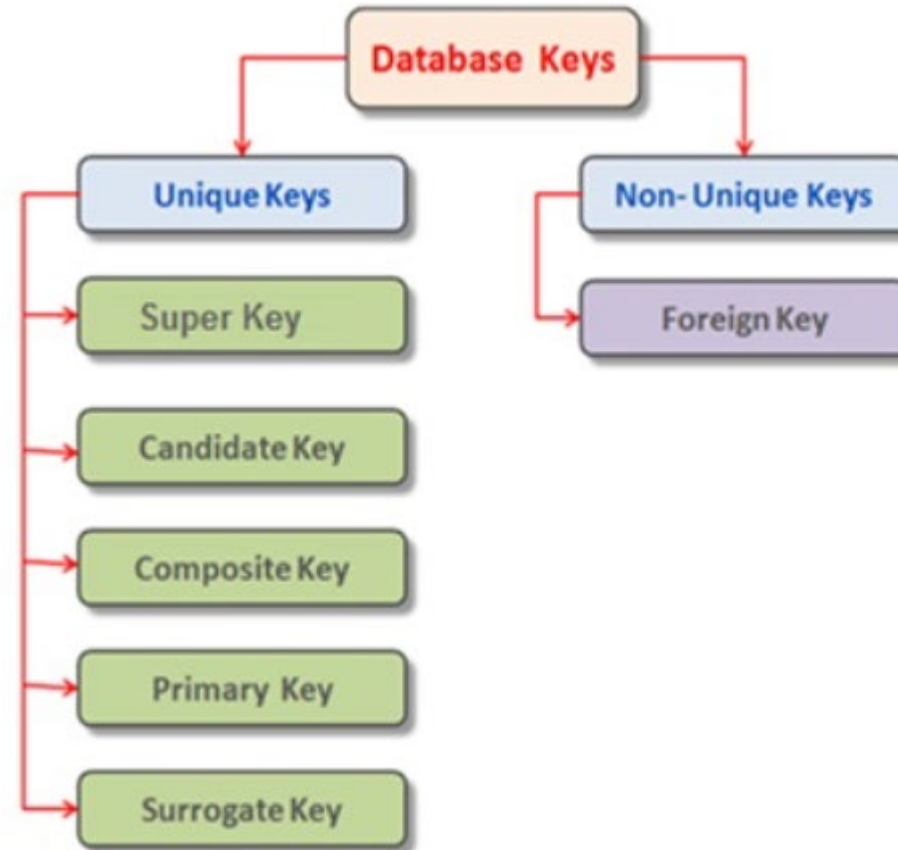
How do we apply a key in Database

In database design phase the database designer **decompose** the larger database tables into the **smaller** tables that contains data about one entity only, then they **analyze all attributes** and find **the one with unique values** to use it as a key attribute.



Relational Database Models

Types of Database Keys



Relational Database Models

SUPER KEY

- Super key is defined as either a **single attribute** or **group of attributes** that can be used to **uniquely identify each row in a table**
- The super key is the **most general form of the unique key**. The table can have many super keys. A set of super key is usually **super-set** of other keys
- It is the **most generic type** of a key

Student Table

Student ID	Student SSN	Student Name
001	2625438	Piter
002	6529754	John
003	5681134	Maya
004	7600912	Kitty

Super Keys Options In The Student Table

{ Student_ID }
{ Student_SSN }
{ Student_ID, Student_SSN }
{ Student_ID, Student_Name }
{ Student_SSN, Student_Name }
{ Student_ID, Student_SSN, Student_Name }

Relational Database Models

SUPER KEY

- The following table have **six number of super-keys**
- The attributes **Student_ID** and **Student_SSN** will **always** have **unique** values for each instance of student table, therefor both of these attributes qualify for the super key.
- The **Student_Name** attribute individually do **not** qualify for **super key**.
- But if use **Student_Name** along with any of super key attribute then it will also become a super key.

Student Table

Student ID	Student SSN	Student Name
001	2625438	Piter
002	6529754	John
003	5681134	Maya
004	7600912	Kitty

Super Keys Options In The Student Table

{ Student_ID }
{ Student_SSN }
{ Student_ID, Student_SSN }
{ Student_ID, Student_Name }
{ Student_SSN, Student_Name }
{ Student_ID, Student_SSN, Student_Name }

Relational Database Models

COMPOSITE KEY

- Any attribute that **cannot uniquely identify the records individually, but It can be unique when combined with any super key** attributes then that attribute will called as composite key
- **For example** in following table **Student_Name** individuly cannot uniquely identify the records. But **when combined with Student_Id or Student_SSN** it can.
- So here **Student_Name** is compsite key.
- A **primary key consisting of more than one attributes** is also called composite key.

Student Table

Student ID	Student SSN	Student Name
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Super Keys Options In The Student Table

{ Student_ID }
{ Student_SSN }
{ Student_ID, Student_SSN }
{ Student_ID, Student_Name }
{ Student_SSN, Student_Name }
{ Student_ID, Student_SSN, Student_Name }

Relational Database Models

CANDIDATE KEY

- Candidate key is defined as the **super key with minimum number of attributes**. In other words it is a super key with **no redundant attributes**.
- Main purpose of candidate key is to **offer multiple options to database designers** so that suitable primary key can be selected.
- Each candidate key has **full ability to function as primary key** but one is selected by database designer.
- Each **candidate key is super key** but every **super key may not be a candidate key**.

Student Table

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Candidate Keys In The Student Table

{ Student_ID }
{ Student_SSN } → Candidate Keys

- Super Keys with minimum number of attributes.
- Super Key without any redundant attribute.

Relational Database Models

PRIMARY KEY

- Primary key is the most important feature of database that is **selected by database Administrator** to uniquely identify all records.
- Each **table** can have only **one primary key**.
- If there are multiple candidate key then one primary key is selected out of them.

Student Table

Student ID	Student SSN	Student Name
001	2625438	Piter
002	6529754	John
003	5681134	Maya
004	7600912	Kitty

Primary Key Options In The Student Table

{ Student_ID }
{ Student_SSN } → Any One Of the Candidate Key

- Each Table Can have only one Primary Key.
- Any One Candidate Key can be selected as Primary Key.

Relational Database Models

ALTERNATE KEY

- In previous slide we have studied that primary key is selected out of the candidate keys
- **For Example:** As we have two candidate keys in following table Student_ID and Student_SSN
- If we have selected Student_ID as primary key in following table
- Then Student_SSN will become Alternate key for the following table.

Student Table

Student ID	Student SSN	Student Name
001	2625438	Piter
002	6529754	John
003	5681134	Maya
004	7600912	Kitty

Alternate Key Options In The Student Table

{ Student_ID } ————— Primary Key

{ Student_SSN } ————— Alternate Key

- Each Table can have one primary key
- After selection of primary key other remanining candidate keys will be known as alternate key

Relational Database Models

FOREIGN KEY

- The concept of **Foreign Key** originates when we want to **create relationship between more than one tables**. It also helps in **creating the parent table and child table relationship**.
- In this case the **primary key of one table is included as an attribute in second table** that attribute will be called as foreign key for the table one.
- The foreign keys helps us in **enforcing referential integrity** that is an important feature of database.
- Referential integrity plays important role in **ensuring the accuracy and consistency** of data.

Student Table

Student ID	First Name	SSN	Course ID
1001	S1	CA245	CS 1
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Instructor ID Is **Primary Key** For Instructor Table And **Foreign Key** In Course Table



Relational Database Models

SURROGATE KEY

- To understand **Surrogate** Key first we need to understand the natural key
- **Natural keys** are basically those attributes who **don't have redundant value** and they **actually belong to the nature and type of entity**. In other words their inclusion helps in identifying the type of entity.
- Now if a **table don't have any natural attribute** with unique values. Then **we add a new attribute to uniquely identify** all attributes as shown in following table.
- That **new attribute will be called as Surrogate Key** which will act like the **primary key**.

Student Count	Student ID	First Name	SSN	Course
1	CS 1001	S1	CA245	Java
2	EE 1002	S2	WA465	Python
3	EE 1003	S3	AR365	Python
4	CS 1002	S4	WA764	Java
5	IE 1005	S5	CA453	C++
6	CS 1003	S6	CA924	Java



Surrogate Key Column Added As Primary Key

